Circuit Analysis (Lab10)

Experiment 9: Verification of Thevenin’s Theorem

Name: ……………………………. Registration No: ………………………...

Date: ……………………………. Grade and Signature: ……………………

<table>
<tr><td>CLO2
CLO3</td><td colspan="4">Use with identification the circuit components, breadboards, multimeter, power supplies, signal generators, and oscilloscopes.
Make measurements in their own constructed electric circuits.</td></tr>
<tr><td>Psychomotor/Affective</td><td>Level1 (1)</td><td>Level2 (2-3)</td><td>Level3 (4-5)</td><td>Level4 (6-7)</td></tr>
<tr><td rowspan="2">Report Marks (3)</td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="4">Total marks (10)</td></tr>
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Objective

The objective of this lab is verify Thevenin’s theorem and to find the full load current.

Apparatus Required

Sr. no	Apparatus	Range	Quantity
1	Regulated Power Supply	Standard	1
2	Signal Generator	Standard	1
3	Oscilloscope	Standard	1
4	Digital Multimeter	Standard	1
5	Breadboard & Wires	Standard	As Required
6	Resistors	1k, 330 ohms	1,3
7	Potentiometer	Standard	1

Description

Statement:

Any linear bilateral, active two terminal network can be replaced by an equivalent voltage source (VTH). Thevenin’s voltage or V OC in series with looking pack resistance RTH.

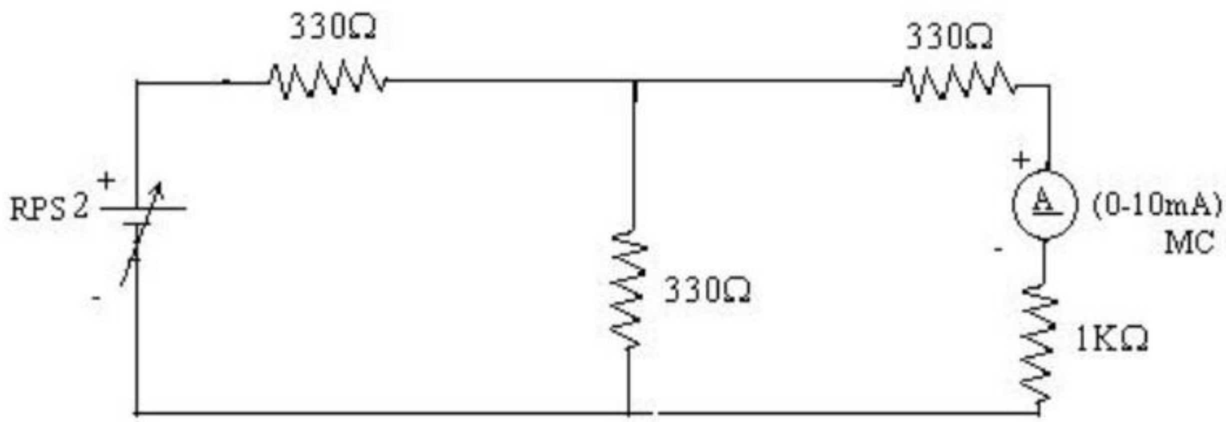
Precautions:

1. Voltage control knob of RPS should be kept at minimum position.
2. Current control knob of RPS should be kept at maximum position

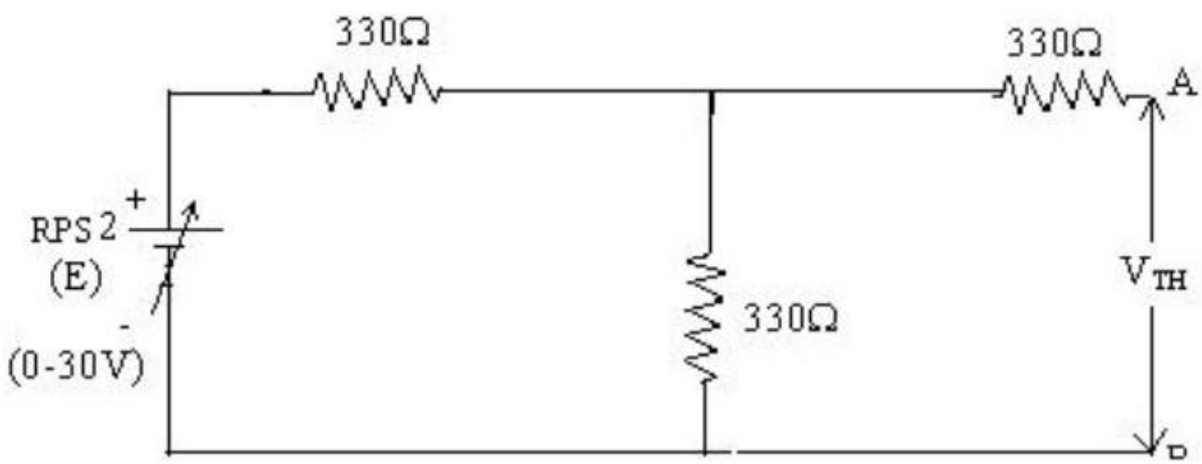
Procedure:

1. Connections are given as per the circuit diagram.
2. Set a particular value of voltage using RPS and note down the corresponding ammeter readings. To find VTH 3. Remove the load resistance and measure the open circuit voltage using multimeter (VTH). To find RTH
4. To find the Thevenin's resistance, remove the RPS and short circuit it and find the RTH using multimeter.
5. Give the connections for equivalent circuit and set VTH and RTH and note the corresponding ammeter reading. 6. Verify Thevenins theorem.

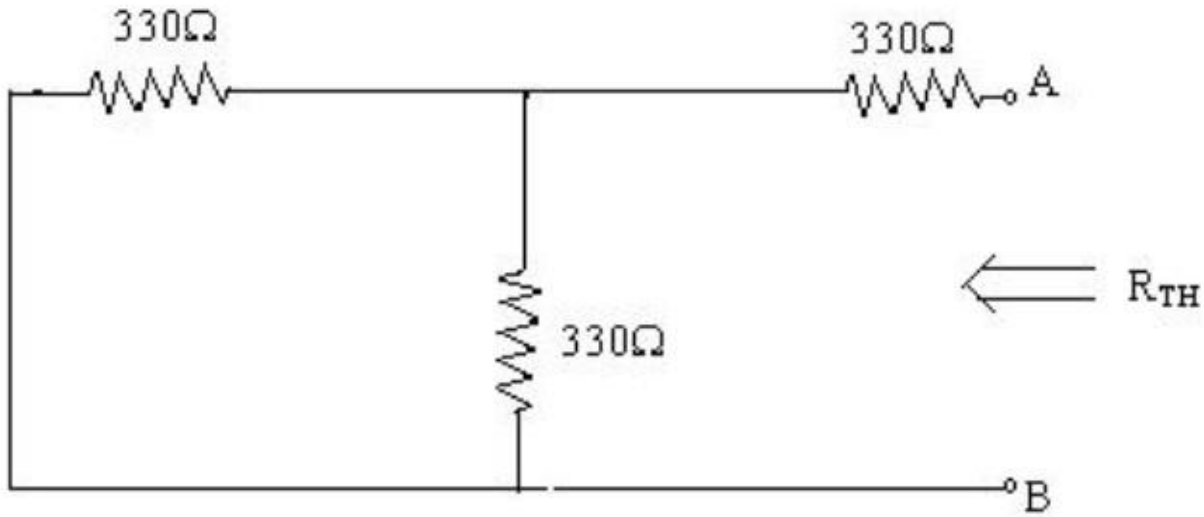
Circuit - 1: To find load current



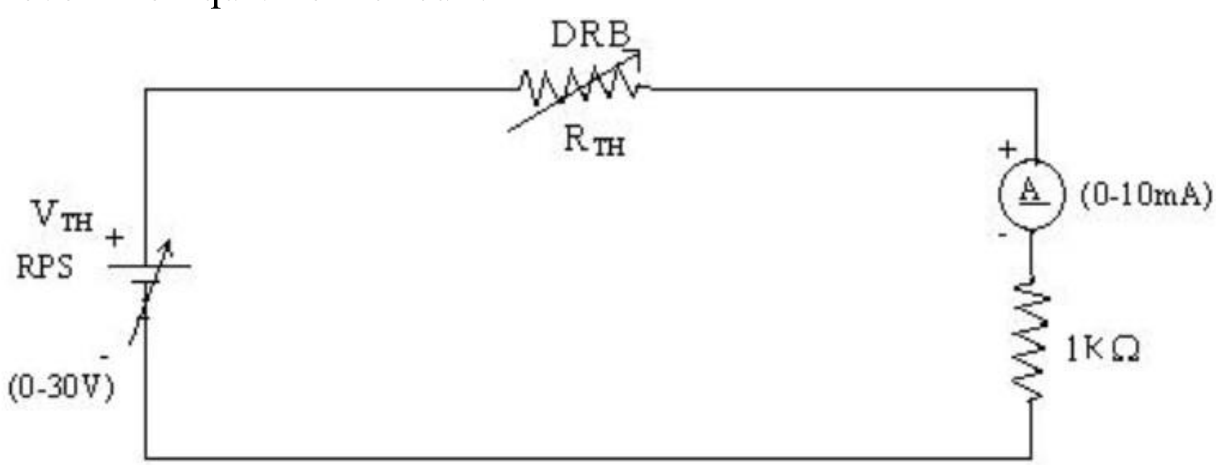
To find V_{TH}



To find R_{TH}



Thevenin's Equivalent circuit:



Thevenin Practical Values

<table>
<tr><td rowspan="2"></td><td rowspan="2">E(V)</td><td rowspan="2">V_{TH}(V)</td><td rowspan="2">R_{TH}(ohms)</td><td colspan="2">I_L (mA)</td></tr>
<tr><td>Circuit - I</td><td>Equivalent Circuit</td></tr>
<tr><td>Values</td><td></td><td></td><td></td><td></td><td></td></tr>
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Post Lab Questions

1. What is Thevenin's Theorem?

2. What are the applications of Thevenin's Theorem?

3. What are the theoretical and practical limitations of Thevenin's Theorem?